

# Solution 6

BS24B006

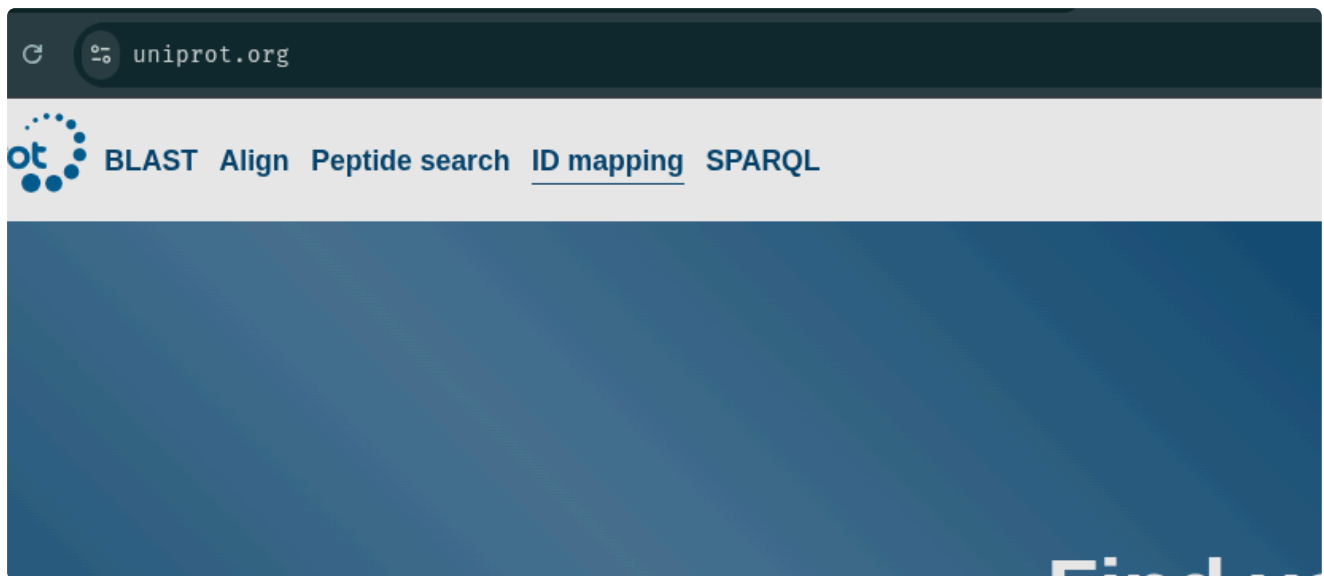
Apoorv Shah

[https://github.com/specapoorv/BT3040\\_BioInfo\\_Lab\\_assignments/tree/main/assignment\\_6](https://github.com/specapoorv/BT3040_BioInfo_Lab_assignments/tree/main/assignment_6)

all the files are in this repo

(note i could not ssh because iitmwifi is very slow today, instead i have clone al2co and did locally instead)

## q1



id mapped the set

A screenshot of the 'Retrieve/ID mapping' form on the UniProt website. The form has a title 'Retrieve/ID mapping' and instructions: 'Enter one or more IDs (100,000 max). You may also load from a text file. Separate IDs by whitespace (space, tab, newline) or commas. This service can also be used programmatically. Alternatively, the underlying data can be downloaded.' A text input field contains the IDs: 'P69905, P01946, P01942, P01966, P01958, P01959, P01965, P06635, P60529, P80043, P01980'. Below the input field, a status bar indicates 'Your input contains 11 unique IDs'. There are two dropdown menus: 'From database' set to 'UniProtKB AC/ID' and 'To database' set to 'UniProtKB'. At the bottom, there is a field to 'Name your ID Mapping job' with the text 'P69905 +10 UniProtKB\_AC-ID -> UniProtKB'.

UniProt

Tools

UniProtKB

Advanced | List

Search



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Help

## Tool results

Your tool analysis results from the last 7 days are listed below. If you have tools jobs running, you can navigate away to other pages and you will be notified once the job is completed.

ID

MAPPING

TPIS\_HUMAN +8 UniProtKB\_AC-ID → UniProtKB

2026-09-15

17:28

Queued

We will notify you when your results are ready

7w9I8540zr

Source database: UniProtKB AC/ID

Target database: UniProtKB



ID

MAPPING

TPIS\_HUMAN +8 UniProtKB\_AC-ID → UniProtKB

2026-09-15

17:27

Completed

7w9I8540zr

Source database: UniProtKB AC/ID

Target database: UniProtKB



ID

MAPPING

P69905 +10 UniProtKB\_AC-ID → UniProtKB

2026-09-15

17:25

Completed

233xR1ma3f

Source database: UniProtKB AC/ID

Target database: UniProtKB



Help

mapped the id given in tutorial to get the fasta sequence (given in the folder of repo as well)

Tools Download (11) Add View: Cards Table Customize columns Resubmit Share 11 rows selected

11 IDs mapped to 11 results

| ✓ From   | Entry  | Entry Name | Protein Names                      | Gene Names   | Organism                                                              | Length |
|----------|--------|------------|------------------------------------|--------------|-----------------------------------------------------------------------|--------|
| ✓ P69905 | P69905 | HBA_HUMAN  | Hemoglobin subunit alpha[...]      | HBA1, HBA2   | Homo sapiens (Human)                                                  | 142 AA |
| ✓ P01946 | P01946 | HBA_RAT    | Hemoglobin subunit alpha-1/2 [...] | Hba1, Hba-a1 | Rattus norvegicus (Rat)                                               | 142 AA |
| ✓ P01942 | P01942 | HBA_MOUSE  | Hemoglobin subunit alpha[...]      | Hba, Hba-a1  | Mus musculus (Mouse)                                                  | 142 AA |
| ✓ P01966 | P01966 | HBA_BOVIN  | Hemoglobin subunit alpha[...]      | HBA          | Bos taurus (Bovine)                                                   | 142 AA |
| ✓ P01958 | P01958 | HBA_HORSE  | Hemoglobin subunit alpha[...]      | HBA          | Equus caballus (Horse)                                                | 142 AA |
| ✓ P01959 | P01959 | HBA_EQUAS  | Hemoglobin subunit alpha[...]      | HBA1, HBA2   | Equus asinus (Donkey) (Equus africanus asinus)                        | 142 AA |
| ✓ P01965 | P01965 | HBA_PIG    | Hemoglobin subunit alpha[...]      | HBA          | Sus scrofa (Pig)                                                      | 141 AA |
| ✓ P06635 | P06635 | HBA_PONPY  | Hemoglobin subunit alpha[...]      | HBA1, HBA2   | Pongo pygmaeus (Bornean orangutan)                                    | 142 AA |
| ✓ P60529 | P60529 | HBA_CANLF  | Hemoglobin subunit alpha[...]      | HBA          | Canis lupus familiaris (Dog) (Canis familiaris)                       | 141 AA |
| ✓ P80043 | P80043 | HBA_TREBE  | Hemoglobin subunit alpha[...]      | hba          | Trematomus bernacchii (Emerald rockcod) (Pseudotrematomus bernacchii) | 142 AA |
| ✓ P01980 | P01980 | HBA_APTFO  | Hemoglobin subunit alpha[...]      | HBA          | Aptenodytes forsteri (Emperor penguin)                                | 142 AA |

UniProt Tools Tool results Advanced | List Search Help

Status  
Reviewed (Swiss-Prot) (9)

Popular organisms  
C. elegans (1)  
Fruit fly (1)  
Human (1)  
Mouse (1)  
S. cerevisiae (1)

Taxonomy  
Filter by taxonomy

Proteins with  
3D structure (6)  
Active site (9)  
Alternative products (isoforms) (4)  
Alternative splicing (4)  
Beta strand (6)  
More items

Protein existence  
Protein level (8)

## ID mapping 9 results found for UniProtKB\_AC-ID → U

Overview Input Parameters API Request

Tools Download (9) Add View: Cards Table Customize columns Resubmit

9 IDs mapped to 9 results

| ✓ From       | Entry  | Entry Name | Protein Names                              | Gene Names                | Organism                 |
|--------------|--------|------------|--------------------------------------------|---------------------------|--------------------------|
| ✓ TPIS_HUMAN | P60174 | TPIS_HUMAN | Triosephosphate isomerase[...]             | TPI1, TPI                 | Homo sapiens (Human)     |
| ✓ TPIS_YEAST | P00942 | TPIS_YEAST | Triosephosphate isomerase[...]             | TPI1, YDR050C, YD9609.05C | Saccharomyces cerevisiae |
| ✓ TPIS_GRAGA | P48492 | TPIS_GRAGA | Triosephosphate isomerase, cytosolic [...] | TPI1                      | Gracilaria (Red algae)   |
| ✓ TPIS_TRYCR | P52270 | TPIS_TRYCR | Triosephosphate isomerase, glycosomal[...] |                           | Trypanosoma              |
| ✓ TPIS_MAIZE | P12863 | TPIS_MAIZE | Triosephosphate isomerase, cytosolic [...] |                           | Zea mays                 |

downloaded in fasta

running clustal omega alignment for both the set

GN=hba PE=1 SV=1  
SLSDKDKAAVRALWSKIGKSADAIGNDALSRMIVVYPOTKTYFSHWPDVTPGSPHIKAHG  
KKVMGGIALAVSKIDDLKTGLMELSEQHAYKL RVDPANFKILNHCI LVVISTMFPKEFTP  
EAHVSLDKFLSGVALALAERYR

Choose file No file chosen

Close

YOUR JOB IS RUNNING

Please note that results can only be retrieved for jobs submitted within the last seven days.

Job ID: clustalo-I20260915-130143-0172-98497456-p1m

RUNNING

Recent Jobs

Close

sequence More example inputs

Parameters

Submit

The EMBL-EBI Job  
tion about this

support staff. If you have  
EBI Support. Read our

Privacy Notice if you are concerned with your privacy and how we handle personal information.

EMBL-EBI is the home for big data in biology.



than the specified value. Otherwise, conservation indices will be set to M-S, where M is the mean conservation value and S is the standard deviation.  
Default = 0.5

- p Input pdb file [File in] Optional  
The sequence in the pdb file should match exactly the first sequence of the alignment.
- d Output pdb file [File Out] Optional  
The B-factors are replaced by the conservation indices.  
Default = STDOUT

```
> ./al2co -i set1.aln -f 0 -c 0 -n F -o set1_method1_entropy_unweighted.txt
No such file: "set1.aln"
Error with reading alignment: 1
> ./al2co -i setA.aln -f 0 -c 0 -n F -o set1_method1_entropy_unweighted.txt
No such file: "setA.aln"
Error with reading alignment: 1
> ls
al2co  al2co.c  GNUmakefile  README
> cd ..
> ls
al2co  BLOSUM62  SetA.aln  SetA.fasta  SetB.aln  SetB.fasta
> ./al2co -i setA.aln -f 0 -c 0 -n F -o set1_method1_entropy_unweighted.txt
zsh: permission denied: ./al2co
> sudo ./al2co -i setA.aln -f 0 -c 0 -n F -o set1_method1_entropy_unweighted.txt
[sudo] password for specapoorv:
sudo: ./al2co: command not found
> sudo al2co/al2co -i setA.aln -f 0 -c 0 -n F -o set1_method1_entropy_unweighted.txt
No such file: "setA.aln"
Error with reading alignment: 1
> sudo al2co/al2co -i SetA.aln -f 0 -c 0 -n F -o set1_method1_entropy_unweighted.txt
> ls
al2co  BLOSUM62  set1_method1_entropy_unweighted.txt  SetA.aln  SetA.fasta  SetB.aln  SetB.fasta
> al2co/al2co -i SetA.aln -f 0 -c 1 -n F -o set1_method2_variance_unweighted.txt
> al2co/al2co -i SetA.aln -f 0 -c 2 -s BLOSUM62 -n F -o set1_method3_sum_of_pairs.txt
> al2co/al2co -i SetA.aln -f 1 -c 1 -n F -o set1_method4_variance_weighted.txt
> al2co/al2co -i SetA.aln -f 0 -c 0 -n T -o set1_method5_entropy_normalized.txt
> al2co/al2co -i SetB.aln -f 0 -c 0 -n F -o set2_method1_entropy_unweighted.txt
Names do not match, was: 3sp|P52270|TPIS_TRYCR, now: sp|P52270|TPIS_TRYCR
Error with reading alignment: 5
> al2co/al2co -i SetB.aln -f 0 -c 0 -n F -o set2_method1_entropy_unweighted.txt
> al2co/al2co -i SetB.aln -f 0 -c 1 -n F -o set2_method2_variance_unweighted.txt
> al2co/al2co -i SetB.aln -f 0 -c 2 -s BLOSUM62 -n F -o set2_method3_sum_of_pairs.txt
> al2co/al2co -i SetB.aln -f 1 -c 1 -n F -o set2_method4_variance_weighted.txt
> al2co/al2co -i SetB.aln -f 0 -c 0 -n T -o set2_method5_entropy_normalized.txt
~/bioinfo/assignment_6 main !1 ?1 >
```

06:11:14 PM

(for clarification, i have cloned the repo and built it locally)

## Q.2

we have to do the same thing except on the web so i will use the same .aln file

## Unweighted frequency and entropy-based measure

### DATA INPUT

Enter protein sequence alignment in [CLUSTAL](#) format:

Clear sequences

Or upload a file  SetA.aln

### DATA SUBMIT

Enter **email** to receive the result (optional):

Enter a **job name** (optional):

### PARAMETERS

- [sequence weighting scheme](#): ☐ henikoff-henikoff ☐ independent count ☒ unweighted
- [conservation calculation method](#): ☒ entropy ☐ variance ☐ sum-of-pairs
- For sum-of-pairs method only:  
[scoring matrix](#):  
☒ BLOSUM62 matrix ☐ identity matrix
- [scoring matrix transformation](#):  
☒ no transformation ☐ normalization ☐ adjustment
- [normalize conservation values](#): ☐ True ☒ False
- [window size used for averaging conservation \(for smoothing purpose\)](#):
- [gap fraction above which conservation calculation is not performed](#):
- [exclude the first sequence from calculation](#): ☐ True ☒ False
- [output alignment block size](#):
- [pdb file for which b-factor field is replaced with conservation \(optional\)](#):  
 No file chosen

[AL2CO documentation](#)

## AL2CO alignment conservation server

Your job is number 1 in queue. Please wait ...  
This page will be updated automatically every 30 seconds until your job is done.  
You can also refresh this page at any time, or bookmark this page and check it later.

### RESULTS:

- The list of positional conservation values is [here](#).
- The alignment with integer conservation indices is [here](#).

### INPUTS:

- Input alignment is [here](#).
- Input pdb file: none.

[http://prodata.swmed.edu/al2co/results/QUERY\\_3PSvl2.csv.txt](http://prodata.swmed.edu/al2co/results/QUERY_3PSvl2.csv.txt)

```

1      :      0.000
2      S      -0.305
3      L      0.000
4      S      0.000
5      D      -1.121
6      K      -1.160
7      D      0.000
8      K      0.000
9      A      -1.169
10     A      -0.305
11     V      -0.586
12     R      -0.305
13     A      -1.034
14     L      -1.160
15     W      -0.305
16     S      -0.916
17     K      0.000
18     I      -0.689
19     G      -0.305

```

same as in the file we did through CLI of al2co (all the files present in github)

## Unweighted frequency and variance-based measure

Or upload a file  SetA.aln

---

**DATA SUBMIT**

Enter **email** to receive the result (optional):

Enter a **job name** (optional):

---

**PARAMETERS**

- [sequence weighting scheme](#): ☐ henikoff-henikoff ☐ independent count ☒ unweighted
- [conservation calculation method](#): ☐ entropy ☒ variance ☐ sum-of-pairs
- For sum-of-pairs method only:  
[scoring matrix](#):  
☒ BLOSUM62 matrix ☐ identity matrix  
[scoring matrix transformation](#):  
☒ no transformation ☐ normalization ☐ adjustment
- [normalize conservation values](#): ☐ True ☒ False
- [window size used for averaging conservation \(for smoothing purpose\)](#):
- [gap fraction above which conservation calculation is not performed](#):
- [exclude the first sequence from calculation](#): ☐ True ☒ False
- [output alignment block size](#):
- [pdb file for which b-factor field is replaced with conservation \(optional\)](#):  
 No file chosen

Unweighted frequency and sum of pairs measure  
(HINT: Download and use BLOSUM62 scoring)

## matrix)

Or upload a file

Choose file

SetA.aln

DATA SUBMIT

Enter **email** to receive the result (optional):

Enter a **job name** (optional):

Submit

Reset

PARAMETERS

- [sequence weighting scheme](#): ☐ henikoff-henikoff ☐ independent count ☒ unweighted
- [conservation calculation method](#): ☐ entropy ☐ variance ☒ sum-of-pairs
- For sum-of-pairs method only:  
[scoring matrix](#):  
☒ BLOSUM62 matrix ☐ identity matrix
- [scoring matrix transformation](#):  
☒ no transformation ☐ normalization ☐ adjustment
- [normalize conservation values](#): ☐ True ☒ False
- [window size used for averaging conservation \(for smoothing purpose\)](#):
- [gap fraction above which conservation calculation is not performed](#):
- [exclude the first sequence from calculation](#): ☐ True ☒ False
- [output alignment block size](#):
- [pdb file for which b-factor field is replaced with conservation \(optional\)](#):  

Choose file

No file chosen

[AL2CO documentation](#)

## Weighted frequency and variance-based measure

Or upload a file  SetA.aln

#### DATA SUBMIT

Enter **email** to receive the result (optional):

Enter a **job name** (optional):

#### PARAMETERS

- [sequence weighting scheme](#): ☒ henikoff-henikoff ☐ independent count ☐ unweighted
- [conservation calculation method](#): ☐ entropy ☒ variance ☐ sum-of-pairs
- For sum-of-pairs method only:  
[scoring matrix](#):  
☒ BLOSUM62 matrix ☐ identity matrix
- [scoring matrix transformation](#):  
☒ no transformation ☐ normalization ☐ adjustment
- [normalize conservation values](#): ☐ True ☒ False
- [window size used for averaging conservation \(for smoothing purpose\)](#):
- [gap fraction above which conservation calculation is not performed](#):
- [exclude the first sequence from calculation](#): ☐ True ☒ False
- [output alignment block size](#):
- [pdb file for which b-factor field is replaced with conservation \(optional\)](#):  
 No file chosen

[AL2CO documentation](#)

**Normalize the scores obtained with (i)**

Or upload a file  SetA.aln

## DATA SUBMIT

Enter **email** to receive the result (optional):

Enter a **job name** (optional):

## PARAMETERS

- [sequence weighting scheme](#): ☐ henikoff-henikoff ☐ independent count ☒ unweighted
- [conservation calculation method](#): ☒ entropy ☐ variance ☐ sum-of-pairs
- For sum-of-pairs method only:  
[scoring matrix](#):  
☒ BLOSUM62 matrix ☐ identity matrix
- [scoring matrix transformation](#):  
☒ no transformation ☐ normalization ☐ adjustment
- [normalize conservation values](#): ☒ True ☐ False
- [window size used for averaging conservation \(for smoothing purpose\)](#):
- [gap fraction above which conservation calculation is not performed](#):
- [exclude the first sequence from calculation](#): ☐ True ☒ False
- [output alignment block size](#):
- [pdb file for which b-factor field is replaced with conservation \(optional\)](#):  
 No file chosen

```
1      :      0.943
2      S      0.253
3      L      0.943
4      S      0.943
5      D     -1.597
6      K     -1.685
7      D      0.943
8      K      0.943
9      A     -1.705
10     A      0.253
11     V     -0.385
12     R      0.253
13     A     -1.399
14     L     -1.685
15     W      0.253
16     S     -1.134
17     K      0.943
18     I     -0.618
19     G      0.253
```

```
assignment_07 = seq1_method5_entropy_1
1      1      -      0.943
2      2      S      0.253
3      3      L      0.943
4      4      S      0.943
5      5      D     -1.597
6      6      K     -1.685
7      7      D      0.943
8      8      K      0.943
9      9      A     -1.705
10     10     A      0.253
11     11     V     -0.385
12     12     R      0.253
13     13     A     -1.399
```

both files show same value and it looks normalised as well with respect to 1

## q.3

```
assignment_6 > get_top_residues.py > ...
5 def parse_al2co_file(filepath):
13     parts = line.split()
14     if len(parts) >= 3 and parts[0].isdigit():
15         pos = int(parts[0])
16         res = parts[1]
17         if "*" in parts or parts[2].endswith("*"):
18             continue
19
20         score_str = parts[2].replace("*", "")
21         try:
22             score = float(score_str)
23             data.append((pos, res, score))
24         except ValueError:
25             continue
26     return data
27
28 # i wrote this
29 def print_top_conservation(data, n=10):
30     sorted_highest = sorted(data, key=lambda x: x[2], reverse=True)[:n]
31     sorted_lowest = sorted(data, key=lambda x: x[2])[:n]
32
33     print(f"Top {n} Highest Conservation Scores")
34     print(f"{'Rank':<6}{'Position':<10}{'Residue':<10}{'Score':<10}")
35     for rank, (pos, res, score) in enumerate(sorted_highest, start=1):
36         print(f"{rank:<6}{pos:<10}{res:<10}{score:<10.3f}")
37
38     print(f"Top {n} Lowest Conservation Scores")
39     print(f"{'Rank':<6}{'Position':<10}{'Residue':<10}{'Score':<10}")
40     for rank, (pos, res, score) in enumerate(sorted_lowest, start=1):
41         print(f"{rank:<6}{pos:<10}{res:<10}{score:<10.3f}")
42
43
44 if __name__ == "__main__":
45     filename = (
46         sys.argv[1]
47         if len(sys.argv) > 1
48         else "/home/specapoorv/bioinfo/assignment_6/set1_method1_entropy_unweighted.txt"
49     )
50     records = parse_al2co_file(filename)
51     print_top_conservation(records, n=10)
```

wrote this thingy (file in github again)

```
FileNotFoundError: [Errno 2] No such file or directory: 'set1_method1_entropy_unweighted
>
> /usr/bin/python3 /home/specapoorv/bioinfo/assignment_6/get_top_residues.py
Top 10 Highest Conservation Scores
Rank  Position  Residue  Score
1      1         -      0.000
2      3         L      0.000
3      4         S      0.000
4      7         D      0.000
5      8         K      0.000
6     17         K      0.000
7     26         G      0.000
8     30         L      0.000
9     32         R      0.000
10    38         P      0.000
Top 10 Lowest Conservation Scores
Rank  Position  Residue  Score
1    117         K     -1.846
2    113         T     -1.720
3     70         L     -1.673
4     73         S     -1.594
5     84         E     -1.414
6    133         G     -1.414
7     36         V     -1.367
8     35         V     -1.295
9      9         A     -1.169
10     6         K     -1.160
~/bioinfo main !1 ?1 > |
```

## Set A

| Rank | Alignment Position | Residue | Conservation Score (Entropy) |
|------|--------------------|---------|------------------------------|
| 1    | 100                | F       | 0.000                        |
| 2    | 101                | K       | 0.000                        |
| 3    | 103                | L       | 0.000                        |
| 4    | 116                | P       | 0.000                        |
| 5    | 121                | P       | 0.000                        |
| 6    | 124                | H       | 0.000                        |
| 7    | 126                | S       | 0.000                        |
| 8    | 127                | L       | 0.000                        |
| 9    | 128                | D       | 0.000                        |
| 10   | 129                | K       | 0.000                        |

| Rank | Alignment Position | Residue | Conservation Score (Entropy) |
|------|--------------------|---------|------------------------------|
| 1    | 117                | K       | -1.846                       |
| 2    | 113                | T       | -1.720                       |
| 3    | 70                 | L       | -1.673                       |
| 4    | 73                 | S       | -1.594                       |
| 5    | 133                | G       | -1.414                       |
| 6    | 84                 | E       | -1.414                       |
| 7    | 36                 | V       | -1.367                       |
| 8    | 35                 | V       | -1.295                       |
| 9    | 9                  | A       | -1.169                       |
| 10   | 14                 | L       | -1.160                       |

## SetB

| Rank | Alignment Position | Residue | Conservation Score (Entropy) |
|------|--------------------|---------|------------------------------|
| 1    | 117                | K       | -1.846                       |
| 2    | 113                | T       | -1.720                       |
| 3    | 70                 | L       | -1.673                       |
| 4    | 73                 | S       | -1.594                       |
| 5    | 133                | G       | -1.414                       |
| 6    | 84                 | E       | -1.414                       |
| 7    | 36                 | V       | -1.367                       |
| 8    | 35                 | V       | -1.295                       |
| 9    | 9                  | A       | -1.169                       |
| 10   | 14                 | L       | -1.160                       |

| Rank | Alignment Position | Residue | Conservation Score (Entropy) |
|------|--------------------|---------|------------------------------|
| 1    | 33                 | T       | -1.889                       |
| 2    | 64                 | Q       | -1.831                       |
| 3    | 147                | A       | -1.831                       |
| 4    | 27                 | E       | -1.735                       |
| 5    | 23                 | V       | -1.677                       |
| 6    | 36                 | H       | -1.677                       |
| 7    | 145                | R       | -1.677                       |
| 8    | 196                | L       | -1.677                       |
| 9    | 199                | R       | -1.677                       |
| 10   | 208                | D       | -1.677                       |

## q.4

core logic is this

```
print(f"{'Pos':<6}{'Entropy':<12}{'Variance':<12}{'SumOfPairs':<12}")

for i in range(aln_len):
    col = [s[i].upper() for s in seqs]

    #ignore gaps to count frequencies
    valid_aas = [aa for aa in col if aa in AMINO_ACIDS]

    if not valid_aas:
        continue

    #unweighted frequencies: count / total_valid
    freqs = {aa: valid_aas.count(aa) / len(valid_aas) for aa in AMINO_ACIDS}

    entropy_score = calc_entropy(freqs)
    variance_score = calc_variance(freqs)
    sop_score = calc_sum_of_pairs(freqs, matrix)

    print(
        f"{i+1:<6}{entropy_score:<12.3}{variance_score:<12.3}{sop_score:<12.3}"
    )
```

we first calculate the unweighted frequencies and pass them into this function

```

def calc_entropy(freqs):
    #C(i) = sum( f_a * ln(f_a) )
    score = 0.0
    for aa in AMINO_ACIDS:
        f = freqs[aa]
        if f > 0:
            score += f * math.log(f)
    return score

def calc_variance(freqs, bg_freq=1.0 / 20.0):
    # C(i) = sqrt( sum( (f_a - f_overall)^2 ) )
    sq_diff_sum = sum((freqs[aa] - bg_freq) ** 2 for aa in AMINO_ACIDS)
    return math.sqrt(sq_diff_sum)

def calc_sum_of_pairs(freqs, matrix):
    # C(i) = sum_a sum_b ( f_a * f_b * S_ab )
    score = 0.0
    for a in AMINO_ACIDS:
        for b in AMINO_ACIDS:
            if freqs[a] > 0 and freqs[b] > 0:
                score += freqs[a] * freqs[b] * matrix.get((a, b), 0)
    return score

```

matrix used here is blosum62 matrix

```

91
92
93 def main():
94     aln_file = "SetA.aln"
95     blosum_file = "BLOSUM62"
96

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

•> cd assignment_6
•>
•> /usr/bin/python3 /home/specapoorv/bioinfo/assignment_6/q4.py

```

| Pos | Entropy | Variance | SumOfPairs |
|-----|---------|----------|------------|
| 1   | 0.0     | 0.975    | 5.0        |
| 2   | -0.305  | 0.886    | 3.01       |
| 3   | 0.0     | 0.975    | 4.0        |
| 4   | 0.0     | 0.975    | 4.0        |
| 5   | -1.12   | 0.582    | 1.15       |
| 6   | -1.16   | 0.623    | 1.16       |
| 7   | 0.0     | 0.975    | 6.0        |
| 8   | 0.0     | 0.975    | 5.0        |
| 9   | -1.17   | 0.567    | 1.87       |
| 10  | -0.305  | 0.886    | 4.66       |
| 11  | -0.586  | 0.744    | 3.6        |
| 12  | -0.305  | 0.886    | 4.5        |
| 13  | -1.03   | 0.636    | 1.91       |
| 14  | -1.16   | 0.623    | 1.52       |
| 15  | -0.305  | 0.886    | 9.31       |
| 16  | -0.916  | 0.623    | 2.26       |
| 17  | 0.0     | 0.975    | 5.0        |
| 18  | -0.689  | 0.674    | 3.5        |
| 19  | -0.305  | 0.886    | 4.99       |
| 20  | -1.03   | 0.636    | 2.4        |
| 21  | -0.886  | 0.71     | 4.37       |
| 22  | -0.474  | 0.808    | 2.88       |
| 23  | -1.03   | 0.636    | 2.24       |
| 24  | -0.908  | 0.649    | 2.45       |
| 25  | -0.886  | 0.71     | 4.31       |
| 26  | 0.0     | 0.975    | 6.0        |
| 27  | -0.886  | 0.71     | 1.83       |
| 28  | -0.305  | 0.886    | 4.51       |
| 29  | -0.305  | 0.886    | 3.35       |

the answer matches with the files from al2co with some slight difference maybe because i did it for only 2 decimals

## q.6

MAFFT < Job Di x +

ebi.ac.uk/jdispatcher/msa/mafft

New Chrome available

Important note: This tool can align up to 500 sequences or a maximum file size of 1 MB.

Input sequence ⓘ

Sequence type

☐ Automatic ☒ Protein ☐ DNA

Close

YOUR JOB IS SUBMITTING

Please note that results can only be retrieved for jobs submitted within the last seven days.

Job ID: mafft-I20260921-165603-0435-94787005-p1m

SUBMITTING

Recent Jobs

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Parameters

Submit

More example inputs

got mafft and muscle .aln files for both the sets

since i used al2co and then parsed them to get top residues i ran al2co again for these 6 files with method 1

```

> ./al2co -i setA-mafft.aln -f 0 -c 0 -n F -o set1_mafft_entropy.txt
zsh: permission denied: ./al2co
> al2co/al2co -i setA-mafft.aln -f 0 -c 0 -n F -o set1_mafft_entropy.txt
No such file: "setA-mafft.aln"
Error with reading alignment: 1
> ls
al2co    q3.py    q5.py    SetA.aln  SetA-mafft.aln  SetB.aln  SetB-mafft.aln
BLOSUM62 q4.py    results  SetA.fasta SetA-Muscle.aln SetB.fasta SetB-Muscle.aln
> ls /dev/input/js*
> al2co/al2co -i SetA-mafft.aln -f 0 -c 0 -n F -o set1_mafft_entropy.txt
wrong len for MVLSGEDKSNIAAWGKIGGHGAEGAEALERMFAFPPTTKTYFPHF-DVSHGSAQVKGH, was: 10, now: 0
Error with reading alignment: 6
~/bioinfo/assignment_6 main !12 ?8 >

```

it gave some error, because mafft on default gives in fasta so i re ran to get it as clustalw format instead

☐ Automatic ☒ Protein ☐ DNA

Paste your sequence here - or use the example sequence

Choose file

SetA.fasta

Use the example

Clear sequence

More example inputs

Parameters

OUTPUT FORMAT ⓘ

ClustalW

More options ▾

Submit

Title

MAFFT's job

Submit



runs correctly now

```
ls /dev/input/js*
al2co/al2co -i SetA-mafft.aln -f 0 -c 0 -n F -o set1_mafft_entropy.txt
wrong len for MVLSGEDKSNIAAWGKIGGHGAEYGAEALERMFASFPPTTKTYFPHF-DVSHGSAQVKGH, was: 10, now: 0
Error with reading alignment: 6
al2co/al2co -i SetA-Muscle.aln -f 0 -c 0 -n F -o set1_muscle_entropy.txt
al2co/al2co -i SetA-mafft.aln -f 0 -c 0 -n F -o set1_mafft_entropy.txt
~/bioinfo/assignment_6 main !12 ?10 >
```

09:46:39 PM

```

> /usr/bin/python3 /home/specapoorv/bioinfo/assignment_6/q5.py
=== (i) Top Residues with SIMILAR Conservation Scores (Diff <= 0.1) ===
Pos  Res  Clustal  MAFFT  MUSCLE  Diff
1    -    0.000    0.000    0.000    0.000
2    S    -0.305   -0.305   -0.305    0.000
3    L    0.000    0.000    0.000    0.000
4    S    0.000    0.000    0.000    0.000
5    D    -1.121   -1.121   -1.121    0.000
6    K    -1.160   -1.160   -1.160    0.000
7    D    0.000    0.000    0.000    0.000
8    K    0.000    0.000    0.000    0.000
9    A    -1.169   -1.169   -1.169    0.000
10   A    -0.305   -0.305   -0.305    0.000

=== (ii) Top Residues with DIFFERENT Conservation Scores (Diff > 0.1) ===
Pos  Res  Clustal  MAFFT  MUSCLE  Diff
>
> /usr/bin/python3 /home/specapoorv/bioinfo/assignment_6/q5.py
=== (i) Top Residues with SIMILAR Conservation Scores (Diff <= 0.1) ===
Pos  Res  Clustal  MAFFT  MUSCLE  Diff
3    S    -0.974   -0.974   -0.974    0.000
4    K    -1.494   -1.494   -1.494    0.000
5    P    -0.377   -0.377   -0.377    0.000
6    Q    -0.736   -0.736   -0.736    0.000
7    P    -0.377   -0.377   -0.377    0.000
8    I    -0.736   -0.736   -0.736    0.000
9    A    -0.377   -0.377   -0.377    0.000
10   A    -0.377   -0.377   -0.377    0.000
11   A    -0.377   -0.377   -0.377    0.000
12   N    0.000    0.000    0.000    0.000

=== (ii) Top Residues with DIFFERENT Conservation Scores (Diff > 0.1) ===
Pos  Res  Clustal  MAFFT  MUSCLE  Diff
126  A    -1.061   -1.061   -0.349    0.712
71   I    -0.849   -0.562   -0.562    0.287
~/bioinfo/main 112 213 >

```

top one is of set1 or setA, bottom is of SetB

so basically For Set 1, all three aligners (Clustal Omega, MAFFT, and MUSCLE) produced the exact same alignment with zero score difference across all residues because the globin sequences share very high sequence similarity with no tricky insertions.

For Set 2, most core residues still matched completely (diff = 0.000), but a few positions like 126 (diff = 0.712) and 71 (diff = 0.287) had different conservation scores. This happens because the sequences in Set 2 are from widely different species with variable loop regions, so MUSCLE placed gaps and shifted columns differently compared to Clustal and MAFFT

## q.6

entropy formula is

$$C(i) = \sum_a f_a \ln(f_a)$$

so we will go at each position count the AAs and calculate frequencies  
and for each residue at that position calculate the entropy

position 9

|                                              |                                                                |     |
|----------------------------------------------|----------------------------------------------------------------|-----|
| TREBE                                        | -SLSDKDIAVVRALWSKIGKSADAIGNDALSRMIVVYPQTKTYFSHWPDVTPGSPHIKAH   | 59  |
| APTFO                                        | MVLSANDISFVKISFSKISSHAEEYGAETLERMFTTYPTKYFPHF-DLHHGSAQVKAH     | 59  |
| CANLF                                        | -VLSPADITIKSTWDKIGGHAGDYGGAEALDRTFQSFPPTKYFPHF-DLSPGSAQVKAH    | 58  |
| RAT                                          | MVLSADDITIKNCWKGKIGGHGGEYGEEALQRMFAAFPTTKTYFSHI-DVSPGSAQVKAH   | 59  |
| PIG                                          | -VLSAADIAFKAAAWGKVGGQAGAHGAEALERMF LGFPPTTKTYFPHF-NLSHGSDQVKAH | 58  |
| MOUSE                                        | MVLSGEDISFIKAAWGKIGGHGAEGAEALERMFASFPTTKTYFPHF-DVSHGSAQVKGH    | 59  |
| PONPY                                        | MVLSPADITIVKTAWGKVGGAHAGDYGAEALERMF LSFPTTKTYFPHF-DLSHGSAQVKDH | 59  |
| HUMAN                                        | MVLSPADITIVKAAWGKVGGAHAGEYGAEALERMF LSFPTTKTYFPHF-DLSHGSAQVKGH | 59  |
| BOVIN                                        | MVLSAADIGFVKAAWGKVGGHAAEYGAEALERMF LSFPTTKTYFPHF-DLSHGSAQVKGH  | 59  |
| HORSE                                        | MVLSAADITIVKAAWSKVGGHAGEYGAEALERMF LGFPPTTKTYFPHF-DLSHGSAQVKAH | 59  |
| EQUAS                                        | MVLSAADITIVKAAWSKVGGNAGEFGAEALERMF LGFPPTTKTYFPHF-DLSHGSAQVKAH | 59  |
| <br>** *:::* . *::* : :* ***** * :: ** ::* * |                                                                |     |
| TREBE                                        | GKKVMGGIALAVSKIDDLKTGLMELSEQHAYKL RVDPANFKILNH CILVVISTMFPEFT  | 119 |

count:

$$A = 2$$
$$S = 2$$
$$T = 6$$
$$G = 1$$

## Frequencies:

- $f_A = \frac{2}{11} \approx 0.1818$
- $f_S = \frac{2}{11} \approx 0.1818$
- $f_T = \frac{6}{11} \approx 0.5455$
- $f_G = \frac{1}{11} \approx 0.0909$

$$f_A \ln(f_A) = 0.1818 \times \ln(0.1818) = 0.1818 \times (-1.7048) \approx -0.3099$$

$$f_S \ln(f_S) = 0.1818 \times \ln(0.1818) = 0.1818 \times (-1.7048) \approx -0.3099$$

$$f_T \ln(f_T) = 0.5455 \times \ln(0.5455) = 0.5455 \times (-0.6061) \approx -0.3306$$

$$f_G \ln(f_G) = 0.0909 \times \ln(0.0909) = 0.0909 \times (-2.3979) \approx -0.2180$$

$$C(9) = -0.3099 - 0.3099 - 0.3306 - 0.2180 = -1.1684 \approx -1.169$$

pos - 11

|                     |             |                                                      |    |
|---------------------|-------------|------------------------------------------------------|----|
| sp P80043 HBA_TREBE | -SLSDKDKAAV | ALWSKIGKSADAIGNDALSRMIVVYPQTKTYFSHWPDVTPGSPHIKAH     | 59 |
| sp P01980 HBA_APTFO | MVLSANDKS   | IVSIFSISKISHAEEYGAETLERMFTTYPQTKTYFPHF-DLHHGSAQVKAH  | 59 |
| sp P60529 HBA_CANLF | -VLSPADKT   | IIISTWDKIGGHAGDYGGGEALDRTFQSFPTTKTYFPHF-DLSPGSAQVKAH | 58 |
| sp P01946 HBA_RAT   | MVLSADDKT   | IINCWGKIGGHGGEYGEEALQRMFAAFPTTKTYFSHI-DVSPGSAQVKAH   | 59 |
| sp P01965 HBA_PIG   | -VLSAADKA   | IVAAWGVGGQAGAHGAEALERMFLGFPTTKTYFPHF-NLSHGSDQVKAH    | 58 |
| sp P01942 HBA_MOUSE | MVLSGEDKS   | IIVAAWGVGGHGAEGYGAELERMFLGFPTTKTYFPHF-DVSHGSAQVKGH   | 59 |
| sp P06635 HBA_PONPY | MVLSPADKT   | IVTAWGKVGAGHAGDYGAEALERMFLSFPTTKTYFPHF-DLSHGSAQVKDH  | 59 |
| sp P69905 HBA_HUMAN | MVLSPADKT   | IVAAWGVGGHAGHAGEYGAEALERMFLSFPTTKTYFPHF-DLSHGSAQVKGH | 59 |
| sp P01966 HBA_BOVIN | MVLSAADKG   | IVAAWGVGGHAAEGYGAELERMFLSFPTTKTYFPHF-DLSHGSAQVKGH    | 59 |
| sp P01958 HBA_HORSE | MVLSAADKT   | IVAAWSKVGGHAGEYGAEALERMFLGFPTTKTYFPHF-DLSHGSAQVKAH   | 59 |
| sp P01959 HBA_EQUAS | MVLSAADKT   | IVAAWSKVGGNAGEFGAEALERMFLGFPTTKTYFPHF-DLSHGSAQVKAH   | 59 |

count:

$$V = 8$$

$$I = 3$$

- Frequencies:**

- $f_V = \frac{8}{11} \approx 0.7273$

- $f_I = \frac{3}{11} \approx 0.2727$

- Entropy calculation:**

-

$$f_V \ln(f_V) = 0.7273 \times \ln(0.7273) = 0.7273 \times (-0.3184) \approx -0.2316$$

$$f_I \ln(f_I) = 0.2727 \times \ln(0.2727) = 0.2727 \times (-1.2993) \approx -0.3543$$

$$C(11) = -0.2316 - 0.3543 = -0.5859 \approx -0.586$$

pos -20

|                     |               |                                                   |    |
|---------------------|---------------|---------------------------------------------------|----|
| sp P80043 HBA_TREBE | -SLSDKDKAAV   | ALWSKIGKSADAIGNDALSRMIVVYPQTKTYFSHWPDVTPGSPHIKAH  | 59 |
| sp P01980 HBA_APTFO | MVLSANDKSNVKS | IFSISKISHAEEYGAETLERMFTTYPQTKTYFPHF-DLHHGSAQVKAH  | 59 |
| sp P60529 HBA_CANLF | -VLSPADKTNIK  | STWDKIGGHAGDYGGGEALDRTFQSFPTTKTYFPHF-DLSPGSAQVKAH | 58 |
| sp P01946 HBA_RAT   | MVLSADDKTNIK  | CWGKIGGHGGEYGEEALQRMFAAFPTTKTYFSHI-DVSPGSAQVKAH   | 59 |
| sp P01965 HBA_PIG   | -VLSAADKANV   | KAAWGVGGQAGAHGAEALERMFLGFPTTKTYFPHF-NLSHGSDQVKAH  | 58 |
| sp P01942 HBA_MOUSE | MVLSGEDKSN    | IKAAWGVGGHGAEGYGAELERMFLGFPTTKTYFPHF-DVSHGSAQVKGH | 59 |
| sp P06635 HBA_PONPY | MVLSPADKTNV   | KTAWGKVGAAGDYGAEALERMFLSFPTTKTYFPHF-DLSHGSAQVKDH  | 59 |
| sp P69905 HBA_HUMAN | MVLSPADKTNV   | KAAWGVGAAGYGAELERMFLSFPTTKTYFPHF-DLSHGSAQVKGH     | 59 |
| sp P01966 HBA_BOVIN | MVLSAADKGNV   | KAAWGVGAAGYGAELERMFLSFPTTKTYFPHF-DLSHGSAQVKGH     | 59 |
| sp P01958 HBA_HORSE | MVLSAADKTNV   | KAAWSKVGAAGYGAELERMFLGFPTTKTYFPHF-DLSHGSAQVKAH    | 59 |
| sp P01959 HBA_EQUAS | MVLSAADKTNV   | KAAWSKVGAAGEFGAEALERMFLGFPTTKTYFPHF-DLSHGSAQVKAH  | 59 |

Count:

$$K = 1$$

S = 1

G = 7

A = 2

- **Frequencies:**

- $f_K = \frac{1}{11} \approx 0.0909$

- $f_S = \frac{1}{11} \approx 0.0909$

- $f_G = \frac{7}{11} \approx 0.6364$

- $f_A = \frac{2}{11} \approx 0.1818$

- **\*\*Entropy calculation:**

$$f_K \ln(f_K) = 0.0909 \times \ln(0.0909) \approx -0.2180$$

$$f_S \ln(f_S) = 0.0909 \times \ln(0.0909) \approx -0.2180$$

$$f_G \ln(f_G) = 0.6364 \times \ln(0.6364) = 0.6364 \times (-0.45198) \approx -0.2876$$

$$f_A \ln(f_A) = 0.1818 \times \ln(0.1818) \approx -0.3099$$

$$C(20) = -0.2180 - 0.2180 - 0.2876 - 0.3099 = -1.0335 \approx -1.034$$

pos-22

|                     |                                                                |    |
|---------------------|----------------------------------------------------------------|----|
| sp P80043 HBA_TREBE | -SLSDKDKAAVRALWSKIGKSAIAIGNDALSRMIVVYPQTKTYFSHWPDVTPGSPHIKAH   | 59 |
| sp P01980 HBA_APTFO | MVLSANDKSNVKSIFSKEISSIAIEYGAETLERMFTTYPQTKTYFPHF-DLHHGSAQVKAH  | 59 |
| sp P60529 HBA_CANLF | -VLSPADKTNIKSTWDKIGGIAADYGGAEALDRTFQSFPPTTKTYFPHF-DLSPGSAQVKAH | 58 |
| sp P01946 HBA_RAT   | MVLSADDKTNKNCWGKIGGIAAGYGEALQRMFAAFPTTKTYFSHI-DVSPGSAQVKAH     | 59 |
| sp P01965 HBA_PIG   | -VLSAADKANVKAAGKVGGAAGAHGAEALERMFLGFPTTKTYFPHF-NLSHGSDQVKAH    | 58 |
| sp P01942 HBA_MOUSE | MVLSGEDKSNIAAGKIGGIAAGYGAELERMFLSFPTTKTYFPHF-DVSHGSAQVKGH      | 59 |
| sp P06635 HBA_PONPY | MVLSPADKTNVKTAWGKVGAIAADYGAELERMFLSFPTTKTYFPHF-DLSHGSAQVKDH    | 59 |
| sp P69905 HBA_HUMAN | MVLSPADKTNVKAAGKVGGAAGYGAELERMFLSFPTTKTYFPHF-DLSHGSAQVKGH      | 59 |
| sp P01966 HBA_BOVIN | MVLSAADKGNVKAAGKVGGAAGYGAELERMFLSFPTTKTYFPHF-DLSHGSAQVKGH      | 59 |
| sp P01958 HBA_HORSE | MVLSAADKTNVKAWSKVGGIAAGYGAELERMFLGFPTTKTYFPHF-DLSHGSAQVKAH     | 59 |
| sp P01959 HBA_EQUAS | MVLSAADKTNVKAWSKVGGIAAGYGAELERMFLGFPTTKTYFPHF-DLSHGSAQVKAH     | 59 |
|                     | ** ** :: :.*. * :.*. : :* ***** * :: ** :.* *                  |    |

count:

A = 9

G = 2

- **Frequencies:**

- $f_A = \frac{9}{11} \approx 0.8182$

- $f_G = \frac{2}{11} \approx 0.1818$

- **Entropy calculation:**

$$f_A \ln(f_A) = 0.8182 \times \ln(0.8182) = 0.8182 \times (-0.2007) \approx -0.1642$$

$$f_G \ln(f_G) = 0.1818 \times \ln(0.1818) = 0.1818 \times (-1.7048) \approx -0.3099$$

$$C(22) = -0.1642 - 0.3099 = -\mathbf{0.4741} \approx -\mathbf{0.474}$$

pos-30

|                     |                                                  |                                |    |
|---------------------|--------------------------------------------------|--------------------------------|----|
| sp P80043 HBA_TREBE | -SLSDKDKAAVRALWSKIGKSADAIGNDALE                  | RMIVVYPQTKTYFSHPDVTGSPHIKAH    | 59 |
| sp P01980 HBA_APTFO | MVLSANDKSNVKSIFSISKISSHAEYGAETL                  | RMFTTYPQTKTYFPHF-DLHHGSAQVKAH  | 59 |
| sp P60529 HBA_CANLF | -VLSPADKTNIKSTWDKIGGHAGDYGGELD                   | RTFQSFPPTTKTYFPHF-DLSPGSAQVKAH | 58 |
| sp P01946 HBA_RAT   | MVLSADDKTNIKNCWKGKIGGHGGEYGEEL                   | CFMFAAFPTTKTYFSHI-DVSPGSAQVKAH | 59 |
| sp P01965 HBA_PIG   | -VLSAADKANVKAAGKVGQAGAHGAEAL                     | RMFLGFPTTKTYFPHF-NLSHGSDQVKAH  | 58 |
| sp P01942 HBA_MOUSE | MVLSGEDKSNIKAAWGKIGGHGAEYGAEL                    | RMFASFPTTKTYFPHF-DVSHGSAQVKGH  | 59 |
| sp P06635 HBA_PONPY | MVLSPADKTNVKTAWGKVGAGHAGDYGAEL                   | RMFLSFPTTKTYFPHF-DLSHGSAQVKDH  | 59 |
| sp P69905 HBA_HUMAN | MVLSPADKTNVKAAGKVGAGHAGEYGAEL                    | RMFLSFPTTKTYFPHF-DLSHGSAQVKGH  | 59 |
| sp P01966 HBA_BOVIN | MVLSAADKGNVKAAGKVGGHAAEYGAEL                     | RMFLSFPTTKTYFPHF-DLSHGSAQVKGH  | 59 |
| sp P01958 HBA_HORSE | MVLSAADKTNVKAWSKVGGHAGEYGAEL                     | RMFLGFPTTKTYFPHF-DLSHGSAQVKAH  | 59 |
| sp P01959 HBA_EQUAS | MVLSAADKTNVKAWSKVGGNAGEFGAEL                     | RMFLGFPTTKTYFPHF-DLSHGSAQVKAH  | 59 |
|                     | ** ** :: :.*. . * : ** : : * ***** * :: ** :.* * |                                |    |

count :

L = 11

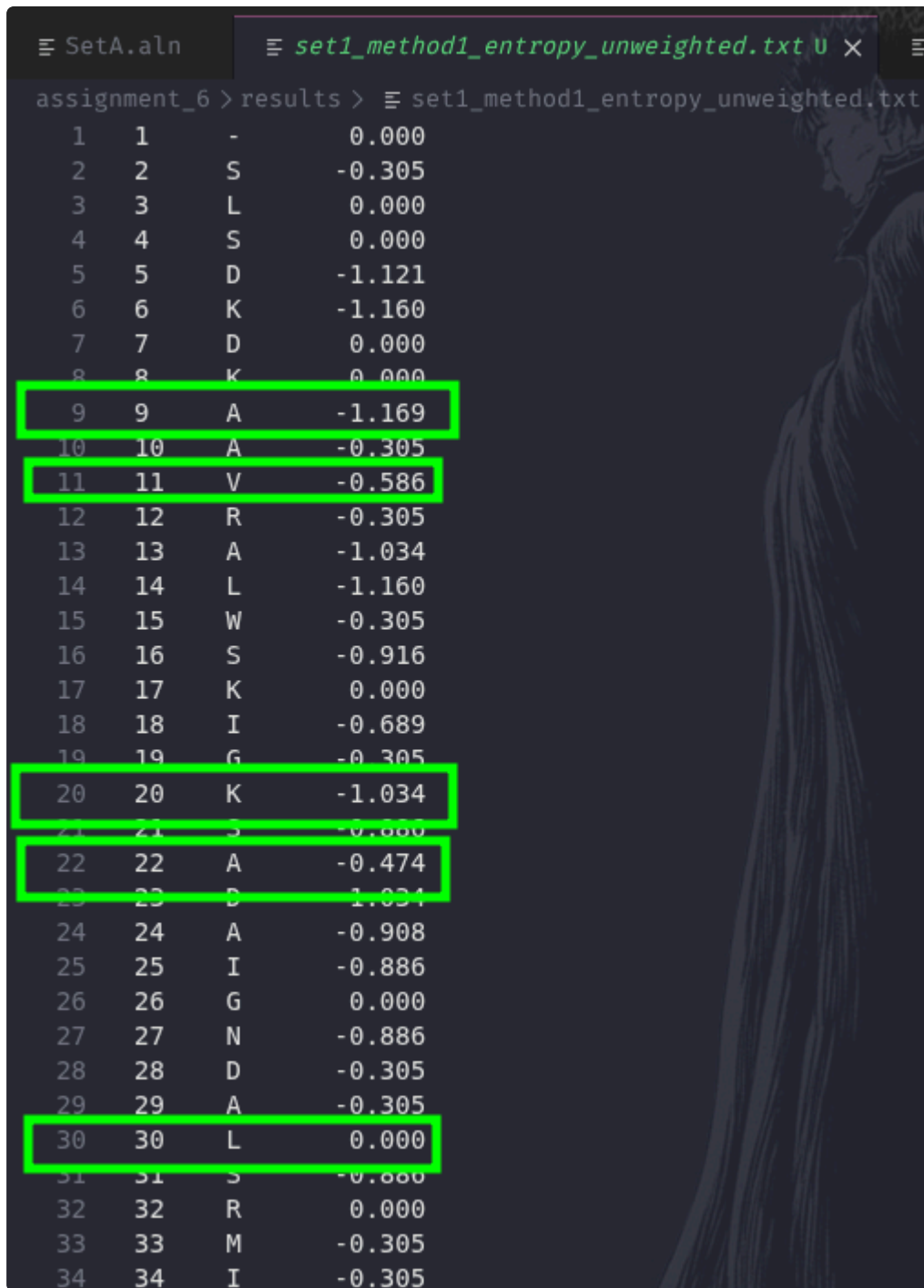
- **Frequencies:**

- $f_L = \frac{11}{11} = 1.0$

- **Entropy calculation:**

$$C(30) = 1.0 \times \ln(1.0) = 1.0 \times 0 = \mathbf{0.000}$$

| Position | Manual Calculated Score | AL2CO Output Score |
|----------|-------------------------|--------------------|
| 9        | -1.169                  | -1.169             |
| 11       | -0.586                  | -0.586             |
| 20       | -1.034                  | -1.034             |
| 22       | -0.474                  | -0.474             |
| 30       | 0.000                   | 0.000              |



assignment\_6 > results > set1\_method1\_entropy\_unweighted.txt

|    |    |   |        |
|----|----|---|--------|
| 1  | 1  | - | 0.000  |
| 2  | 2  | S | -0.305 |
| 3  | 3  | L | 0.000  |
| 4  | 4  | S | 0.000  |
| 5  | 5  | D | -1.121 |
| 6  | 6  | K | -1.160 |
| 7  | 7  | D | 0.000  |
| 8  | 8  | K | 0.000  |
| 9  | 9  | A | -1.169 |
| 10 | 10 | A | -0.305 |
| 11 | 11 | V | -0.586 |
| 12 | 12 | R | -0.305 |
| 13 | 13 | A | -1.034 |
| 14 | 14 | L | -1.160 |
| 15 | 15 | W | -0.305 |
| 16 | 16 | S | -0.916 |
| 17 | 17 | K | 0.000  |
| 18 | 18 | I | -0.689 |
| 19 | 19 | G | -0.305 |
| 20 | 20 | K | -1.034 |
| 21 | 21 | S | -0.886 |
| 22 | 22 | A | -0.474 |
| 23 | 23 | D | -1.034 |
| 24 | 24 | A | -0.908 |
| 25 | 25 | I | -0.886 |
| 26 | 26 | G | 0.000  |
| 27 | 27 | N | -0.886 |
| 28 | 28 | D | -0.305 |
| 29 | 29 | A | -0.305 |
| 30 | 30 | L | 0.000  |
| 31 | 31 | S | -0.886 |
| 32 | 32 | R | 0.000  |
| 33 | 33 | M | -0.305 |
| 34 | 34 | I | -0.305 |

q.7

☒ Amino Acids ☐ Nucleotides

Is there a known structure? 

Insert PDB/UniProt ID

1BTM

or upload a structure 

Select chain identifier

Chain A

☐

No, there is no known structure

Insert a sequence

Paste your sequence here fasta format only

Or, upload sequence File 

If you want to predict the 3D structure using MODELLER, insert MODELLER key 

Job name *\*required*

1BTM\_A

Your email address *\*required*

apoorvshah024@gmail.com

Submit With Default Parameters

Clear

Select Run Parameters Manually

submitted with default parameters, i dont understand most of the parameters listed there



ConSurf Job Status Page - **FINISHED**

[Go To The Results](#)

Job title

1BTM\_A

Date

21/09/2026

## Running Parameters

### Structure

PDB ID: 1BTM

Chain identifier: A

### Phylogenetic Progress

Neighbor Joining with ML  
distance

### Alignment

Multiple Sequence Alignment is  
built using MAFFT

The Homologues are collected  
from UNIREF90

Homolog search algorithm:  
HMMER

HMMER E-value: 0.0001

No. of HMMER Iterations: 1

Maximal %ID Between  
Sequences : 95

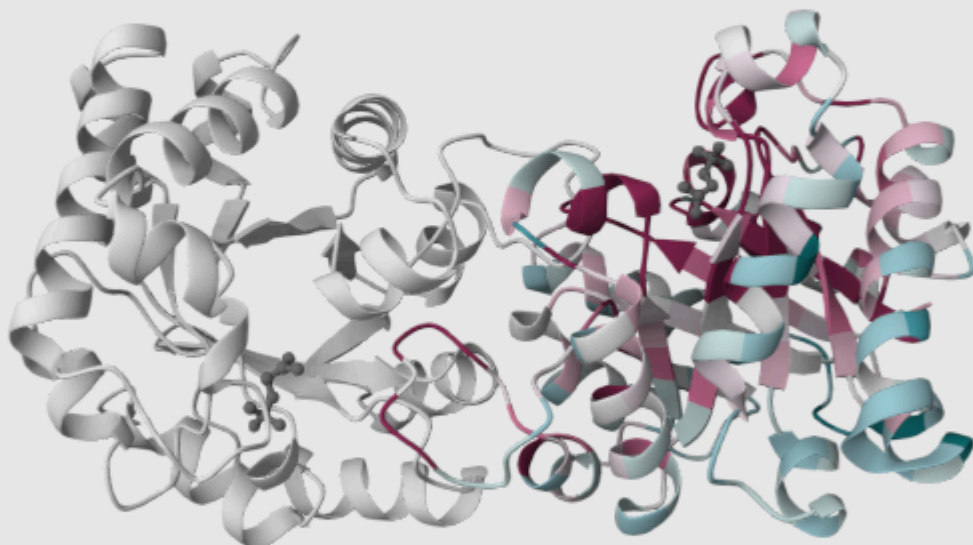
Minimal %ID For Homologs : 35

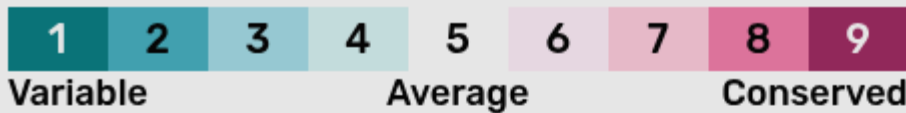
Sequence of 1BTM Chain 1: TRIOSEPH... A

1 RKPIIAGNWKMHKTLAEAVQFVEDVKGHVPPADEVISVVCAPFLFLDRLVQAADGTDLKIGAQTMHFADQGAYTGEVSP  
81 11 21 31 41 51 61 71  
VMLKDLGVTYVILGHSERRQMFAETDETVNKKVLAATRGLIPIICCGESLEEREAGQTNAVVASQVEKALAGLTPEQV  
161 91 101 111 121 131 141 151  
KQAVIAYEPIWAIGTGSSTPEDANSVCGHIRSVVSRLFGPEAAEAIQYGGSVKPDNIRDFLAQQQIDGPLVGGASL  
241 251



[1/1] 21/09/2026, 22:47:07



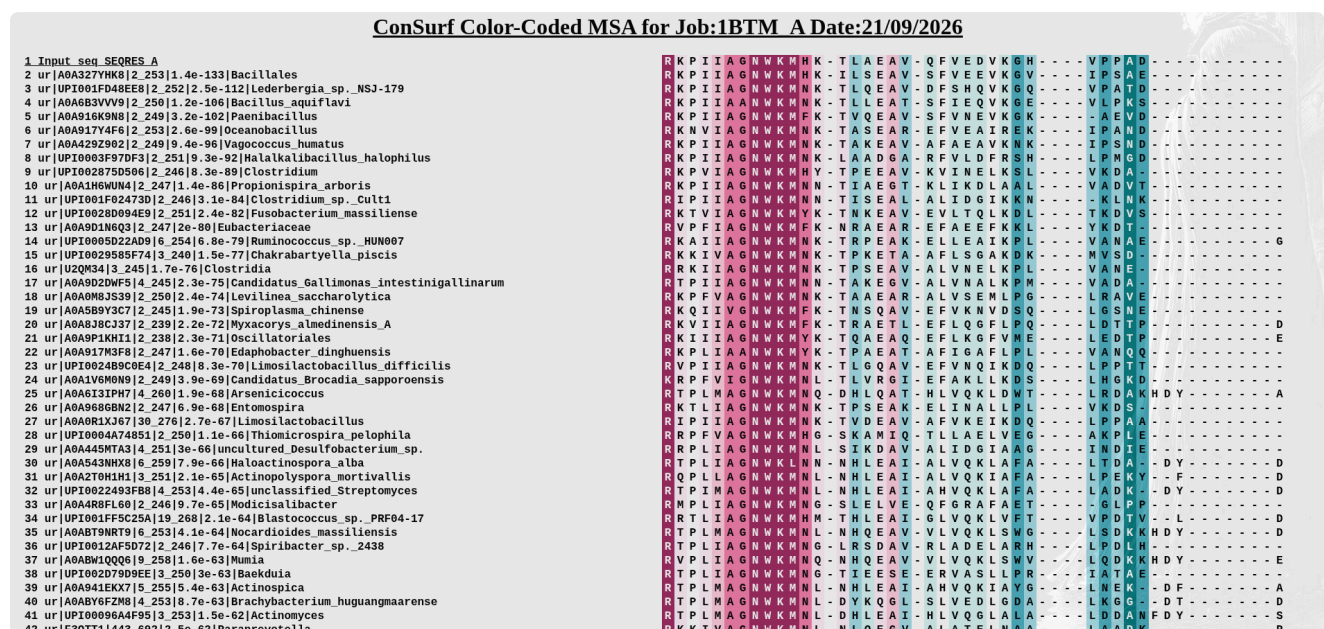


Insufficient Data

- [Download Scale](#)



this site gives lot of information they used neural nets as well



downloaded and added in the my repo (github)  
named as `1BTM_A_consurf_grades.txt`

| POS | SEQ | ATOM     | SCORE  | COLOR | CONFIDENCE | INTERVAL   | B/E | F/S | MSA DATA | RESIDUE VARIETY |
|-----|-----|----------|--------|-------|------------|------------|-----|-----|----------|-----------------|
| 1   | R   | ARG:1:A  | -1.276 | 9     | -1.331,    | -1.249 9,9 | e   | f   | 89/150   | R 98%, K 1%     |
| 2   | K   | LYS:2:A  | -0.239 | 6     | -0.459,    | -0.071 7,5 | e   |     | 102/150  | K 42%, R 29%, T |
| 3   | P   | PRO:3:A  | 0.195  | 5     | -0.071,    | 0.373 5,4  | e   |     | 117/150  | P 52%, K 20%, T |
| 4   | I   | ILE:4:A  | -0.263 | 6     | -0.459,    | -0.183 7,6 | e   |     | 132/150  | L 47%, I 27%, F |
| 5   | I   | ILE:5:A  | -0.554 | 7     | -0.734,    | -0.459 7,7 | b   |     | 135/150  | I 49%, V 32%, M |
| 6   | A   | ALA:6:A  | -0.751 | 8     | -0.897,    | -0.673 8,7 | e   | f   | 141/150  | A 67%, G 12%, V |
| 7   | G   | GLY:7:A  | -0.968 | 8     | -1.075,    | -0.897 9,8 | b   |     | 141/150  | G 87%, A 7%, T  |
| 8   | N   | ASN:8:A  | -1.299 | 9     | -1.331,    | -1.279 9,9 | e   | f   | 141/150  | N 95%, S 4%, T  |
| 9   | W   | TRP:9:A  | -1.168 | 9     | -1.249,    | -1.113 9,9 | b   | s   | 141/150  | W 95%, F 3%, M  |
| 10  | K   | LYS:10:A | -1.322 | 9     | -1.341,    | -1.306 9,9 | e   | f   | 142/150  | K               |
| 11  | M   | MET:11:A | -1.262 | 9     | -1.306,    | -1.249 9,9 | e   | f   | 143/150  | M 96%, L 2%, A  |
| 12  | H   | HIS:12:A | -0.885 | 8     | -0.990,    | -0.846 8,8 | e   | f   | 143/150  | N 77%, H 13%, Y |
| 13  | K   | LYS:13:A | -0.358 | 6     | -0.536,    | -0.285 7,6 | e   |     | 143/150  | G 45%, K 24%, L |
| 14  | T   | THR:14:A | -0.159 | 6     | -0.376,    | -0.071 6,5 | e   |     | 143/150  | T 39%, S 16%, N |
| 15  | L   | LEU:15:A | 1.115  | 3     | 0.579,     | 1.191 4,3  | e   |     | 143/150  | L 24%, R 16%, H |
| 16  | A   | ALA:16:A | 0.544  | 4     | 0.203,     | 0.579 5,4  | e   |     | 143/150  | A 34%, L 11%, S |
| 17  | E   | GLU:17:A | -0.210 | 6     | -0.376,    | -0.071 6,5 | e   |     | 142/150  | E 42%, S 15%, A |
| 18  | A   | ALA:18:A | -0.530 | 7     | -0.673,    | -0.459 7,7 | b   |     | 142/150  | A 44%, L 14%, M |
| 19  | V   | VAL:19:A | 1.065  | 3     | 0.579,     | 1.191 4,3  | e   |     | 142/150  | A 20%, V 12%, I |
| 20  | Q   | GLN:20:A | 0.893  | 4     | 0.373,     | 1.191 4,3  | e   |     | 143/150  | E 29%, A 26%, Q |
| 21  | F   | PHE:21:A | -0.120 | 5     | -0.376,    | 0.056 6,5  | b   |     | 144/150  | L 54%, F 20%, A |
| 22  | V   | VAL:22:A | 0.459  | 4     | 0.203,     | 0.579 5,4  | b   |     | 144/150  | V 29%, L 20%, I |
| 23  | E   | GLU:23:A | 0.568  | 4     | 0.203,     | 0.840 5,4  | e   |     | 145/150  | A 21%, Q 17%, E |
| 24  | D   | ASP:24:A | 1.056  | 4     | 0.579,     | 1.191 4,3  | e   |     | 145/150  | G 17%, E 15%, K |
| 25  | V   | VAL:25:A | 0.056  | 5     | -0.183,    | 0.203 6,5  | b   |     | 145/150  | L 46%, I 17%, V |
| 26  | K   | LYS:26:A | 0.712  | 4     | 0.373,     | 0.840 4,4  | e   |     | 145/150  | A 26%, K 20%, L |
| 27  | G   | GLY:27:A | 2.033  | 2     | 1.191,     | 1.732 3,3  | e   |     | 145/150  | A 18%, S 11%, D |
| 28  | H   | HIS:28:A | 1.356  | 3     | 0.840,     | 1.732 4,3  | e   |     | 145/150  | A 24%, G 15%, I |
| 29  | V   | VAL:29:A | 1.111  | 3     | 0.579,     | 1.191 4,3  | b   |     | 103/150  | L 40%, V 19%, I |
| 30  | P   | PRO:30:A | 1.821  | 2     | 1.191,     | 1.732 3,3  | e   |     | 136/150  | P 16%, A 15%, K |
| 31  | P   | PRO:31:A | 1.650  | 3     | 1.191,     | 1.732 3,3  | e   |     | 145/150  | D 21%, L 15%, K |